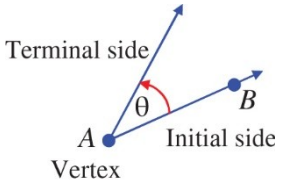
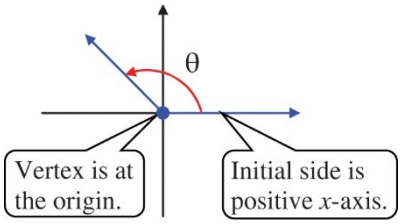
Find Degree Measure

An **angle** is formed by rotating a ray about its endpoint.
Initial side of the angle is the starting position of the ray.
Terminal side of the angle is the final position of the ray.
Vertex of the angle is the common endpoint, often named by a capital letter such as A.

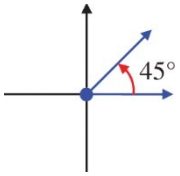




An angle is in **standard position** if its vertex is at the origin in the xy-plane, and its initial side is the positive x-axis.

The **measure** of an angle quantifies the direction and amount of rotation from the initial side to the terminal side.

The measure of an angle is *positive* if the rotation is counterclockwise.
The measure of an angle is *negative* if the rotation is clockwise.

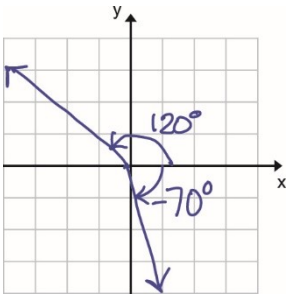


A **degree** is one way to measure an angle.
One full rotation of a ray about its endpoint is 360 degrees, denoted 360°.

Therefore, 1° is $\frac{1}{360}$ of a full rotation.

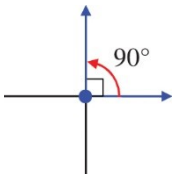
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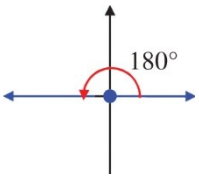
1. Sketch the angles in standard position: a. 120° Q II b. −70° Q IV

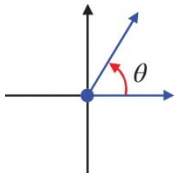


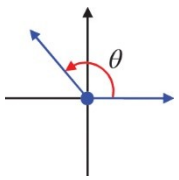
Named Angles:

Right angle	Straight angle	Acute angle	Obtuse angle
Measures 90° (one-quarter turn)	Measures 180° (one-half turn)	$(0° < \theta < 90°)$	$(90° < \theta < 180°)$









Complementary angles:
sum of the measures of two angles is 90°

The *complement* of a 30° angle is a __60__° angle.

Supplementary angles:

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sum of the measures of two angles is 180°

The *supplement* of a 130° angle is a __50__° angle.

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A degree can be divided into 60 equal parts called minutes (min or ′).

1 degree = 60 minutes or 1 minute = $\frac{1}{60}$ degree

Each minute is divided into 60 equal parts called seconds (sec or ″).

1 minute = 60 seconds, 1 degree = 3600 seconds or 1 second = $\frac{1}{60}$ minute = $\frac{1}{3600}$ degree

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2. Convert to decimal degrees: $38°12'$

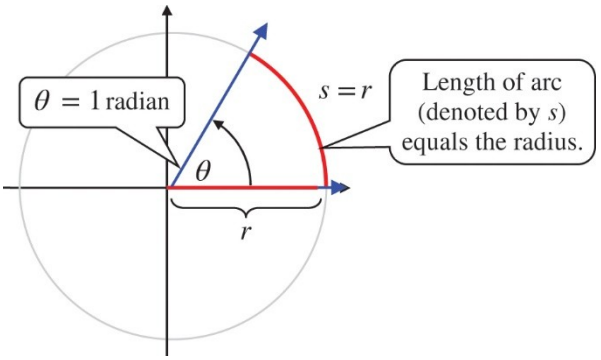
$$38° + \cancel{12}^{1}\ \cancel{\text{min}} \cdot \frac{1°}{\cancel{60}_{5}\ \cancel{\text{min}}} = 38° + 0.2° = 38.2°$$

3. Convert to degrees, minutes, seconds (DMS): 120.45°

$$120° + 0.45\cancel{°} \cdot \frac{60'}{1\cancel{°}} = 120° + 27' = 120°\ 27'$$

Find Radian Measure

When two lines or rays cross a circle, the part of the circle between the intersection points is called the *intercepted arc* and is often denoted by *s*. A *central angle* is an angle with the vertex at the center of a circle.

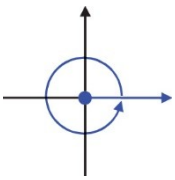


A central angle that intercepts an arc on the circle with length equal to the radius of the circle has a measure of **1 radian.**

1 radian is simply written as 1. Radian measure carries no units.

Radian Measure of an Angle: The radian measure of a central angle θ subtended by an arc of length s on a circle of radius r is given by $\theta = \frac{s}{r}$.

Number of radians in one revolution: $\theta = \frac{s}{r} = \frac{\text{circumference of circle}}{\text{radius of circle}} = \frac{2\pi r}{r} = 2\pi \approx 6.28$



The measure of one full revolution in degrees is 360°. Thus we have $2\pi = 360°$ or $\pi = 180°$.

Converting Between Degrees and Radians: degrees to radians, multiply the degree measure by $\frac{\pi}{180°}$

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radians to degrees, multiply the radian measure by $\frac{180°}{\pi}$

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4. Convert from degrees to radians.

a. $\cancel{120^\circ}^{2} \cdot \frac{\pi}{\cancel{180^\circ}_{3}} = \frac{2\pi}{3}$

b. $-\cancel{300^\circ}^{5} \cdot \frac{\pi}{\cancel{180^\circ}_{3}} = \frac{-5\pi}{3}$

5. Convert from radians to degrees.

a. $\frac{8\cancel{\pi}}{\cancel{9}_{1}} \cdot \frac{\cancel{180^\circ}^{20}}{\cancel{\pi}} = 160^\circ$

b. $-1.8 \cdot \frac{180^\circ}{\pi} = \frac{-324^\circ}{\pi}$

$\approx -103.13^\circ$

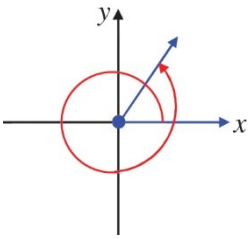
Determine Coterminal Angles

Coterminal Angles: Two angles in standard position with the same initial side and same terminal side.

Two angles in standard position are coterminal if their measures differ by a multiple of 360° or 2π radians.

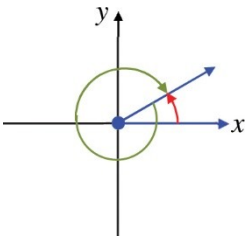
$\theta = 50^\circ$ and $\theta = 410^\circ$

$50^\circ + 360^\circ = 410^\circ$



$\theta = \frac{\pi}{6}$ and $\theta = -\frac{11\pi}{6}$

$\frac{\pi}{6} - 2\pi = -\frac{11\pi}{6}$



---

6. Find a positive angle and a negative angle that are coterminal to the given angle.

a. 110°

$110^\circ + 360^\circ = 470^\circ$

$110^\circ - 360^\circ = -250^\circ$

b. $-\frac{\pi}{4}$

$-\frac{\pi}{4} + 2\pi = -\frac{\pi}{4} + \frac{8\pi}{4} = \frac{7\pi}{4}$

$-\frac{\pi}{4} - 2\pi = -\frac{\pi}{4} - \frac{8\pi}{4} = -\frac{9\pi}{4}$

7. Find an angle between 0° and 360° that is coterminal to the given angle.

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a. $910°$

$$\begin{array}{r} 910° \\ -360° \\ \hline 550° \\ -360° \\ \hline 190° \end{array}$$

or

$$\begin{aligned} 910° - 2(360°) \\ = 190° \end{aligned}$$

b.
$$\begin{aligned} -80° \\ -80° + 360° = 280° \end{aligned}$$

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8. Find an angle between 0 and 2π that is coterminal to the given angle.

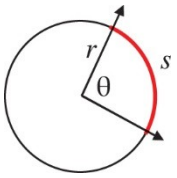
a. $\dfrac{17\pi}{6} - 2\pi = \dfrac{17\pi}{6} - \dfrac{12\pi}{6} = \dfrac{5\pi}{6}$

b. $-\dfrac{35\pi}{12} + 2\pi = \dfrac{-35\pi}{12} + \dfrac{24\pi}{12} = \dfrac{-11\pi}{12}$

$\dfrac{-11\pi}{12} + \dfrac{24\pi}{12} = \dfrac{13\pi}{12}$

Compute Arc Length of a Sector of a Circle

Arc Length: Given a circle of radius r, the length s of an arc intercepted by a central angle θ (in radians) is given by $s = r\theta$.



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9. Find the exact length of the arc intercepted by a central angle θ on a circle of radius r. Then round to the nearest tenth of a unit.

a. $\theta = \dfrac{\pi}{12}, r = 14$ m

$s = r\theta = \left(\cancel{14}^{7}\,m\right)\left(\dfrac{\pi}{\cancel{12}_{6}}\right)$

$= \dfrac{7}{6}\pi \text{ m} \approx 3.7 \text{ m}$

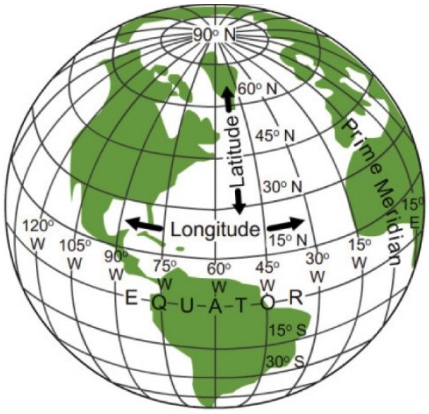
b. $\theta = 20°, r = 2$ ft

$\theta = \cancel{20}^{1}° \cdot \dfrac{\pi}{\cancel{180}_{9}°} = \dfrac{\pi}{9}$

$s = r\theta$

$= (2\text{ft})\left(\dfrac{\pi}{9}\right)$

$= \dfrac{2\pi}{9}\text{ ft} \approx 0.7 \text{ ft}$



10. Key West, Florida is located at 24.1°N, 81.8°W, and Cleveland, Ohio is located at 41.5°N, 81.7°W. Since the longitudes are nearly the same, the cities are roughly due north-south of each other. Using the difference in latitude, approximate the distance between the cities assuming that the radius of the earth is 3960 mi. Round to the nearest mile.

Key West	24.1°N	81.8°W
Cleveland	41.5°N	81.7°W

$41.5° - 24.1° = 17.4°$

$\theta = 17.4° \cdot \dfrac{\pi}{180°} \approx 0.30 \text{ (radians)}$

$s = r\theta$

$= (3960 \text{ mi})(0.30) \approx 1188 \text{ miles}$

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Compute Linear and Angular Speed

Angular and Linear Speed: If a point on a circle of radius r moves through an angle of θ radians in time t, the angular and linear speeds of the point are

angular speed: $\omega = \dfrac{\theta}{t}$ linear speed: $v = \dfrac{s}{t}$ or $v = \dfrac{r\theta}{t}$ or $v = r\omega$.

$d = 1 \rightarrow r = 0.5$

11. A wheel with a diameter of 1 foot spins at a rate of 40 rpm.

a. What is the angular speed?

$$\omega = \frac{\theta}{t} = \frac{40 \text{ rev}}{1 \text{ min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}}$$

$$= 80\pi \frac{\text{rad}}{\text{min}} \quad \text{or} \quad \frac{80\pi}{\text{min}}$$

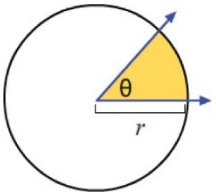
b. How fast in feet per minute does a point on the edge of the wheel spin?

$$v = r\omega$$

$$= (0.5\text{ft})\left(\frac{80\pi}{\text{min}}\right)$$

$$= 40\pi \frac{\text{ft}}{\text{min}} \approx 126 \frac{\text{ft}}{\text{min}}$$

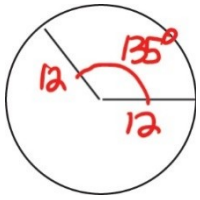
Compute the Area of a Sector of a Circle



The area A of a sector of a circle of radius r with central angle θ (in radians) is given by

$$A = \frac{1}{2} r^2 \theta$$

12. A circle has a radius of 12 feet. Find the area of a sector made by an angle of 135°. Find the exact area of the sector. Then round the result to the nearest tenth of a unit.



$$\theta = \cancel{135^\circ}^{3} \cdot \frac{\pi}{\cancel{180^\circ}_{4}} = \frac{3\pi}{4}$$

$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2}(12\text{ft})^2 \left(\frac{3\pi}{4}\right)$$

$$= \frac{1}{2} \cdot 144\text{ft}^2 \cdot \frac{3\pi}{4}$$

$$= 54\pi\,\text{ft}^2 \approx 169.6\,\text{ft}^2$$

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